

**Chapter 2 Review**

Name: \_\_\_\_\_ Period: \_\_\_\_\_

**2.1 Conditional Statements**

- 1.**
- Rewrite each statement in if-then form.

a) All right angles are congruent.  
\_\_\_\_\_b) I go to practice whenever school is in session.  
\_\_\_\_\_

- 2.**
- Identify the hypothesis and the conclusion: If a polygon has five sides, then it is a pentagon.

Hypothesis: \_\_\_\_\_ Conclusion: \_\_\_\_\_

- 3.**
- Write the converse, inverse, and contrapositive of "If a number is even, then it is divisible by 2." Then tell whether each is true or false.

Converse:

Inverse:

Contrapositive:

Since the conditional and its converse are both true, write the biconditional:  
\_\_\_\_\_

- 4.**
- Write the negation.

a) The angle is obtuse.

b) The lines are not parallel.

- 5.**
- Can "If a number is divisible by 5, then it ends in 0" be written as a biconditional? Explain.

- 6.**
- Use the diagram. Decide whether each statement is true or false. Explain.

a)  $\angle AMC$  and  $\angle CMB$  form a linear pair.

TRUE / FALSE because

b)  $\angle AMC \cong \angle BMD$ 

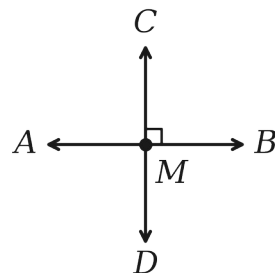
TRUE / FALSE because

c) Ray MA and ray MC are opposite rays.

TRUE / FALSE because

d)  $\angle AMD$  is acute.

TRUE / FALSE because



## 2.2 Inductive & Deductive Reasoning

7. Find the next two items in each pattern.

a) 3, 7, 11, 15, ... \_\_\_\_\_, \_\_\_\_\_  
4, ... \_\_\_\_\_, \_\_\_\_\_

b) 1, 2, 4, 8, ... \_\_\_\_\_, \_\_\_\_\_

c) -5, -2, 1,

8. Find a counterexample for each conjecture.

a) The square of a number is always greater than the number.

b) If two angles are supplementary, then they form a linear pair.

c) Every quadrilateral has four right angles.

9. Inductive or deductive? Circle one and explain.

a) Every time Coach blows the whistle twice, practice ends. He just blew it twice, so practice is over.

INDUCTIVE / DEDUCTIVE because

b) All squares are rectangles. ABCD is a square. So ABCD is a rectangle.

INDUCTIVE / DEDUCTIVE because

10. Is each conjecture valid? If valid, name the law. If not, explain.

a) If a bird is a penguin, then it cannot fly. Pete is a penguin. Conjecture: Pete cannot fly.

b) If  $x$  is a multiple of 8, then  $x$  is even. The number 14 is even. Conjecture: 14 is a multiple of 8.

c) If a boy is in Room 406, then he is in Honors Geometry. If a boy is in Honors Geometry, then he writes proofs. Conjecture: If a boy is in Room 406, then he writes proofs.

## 2.3 Postulates & Diagrams

11. Name the postulate.

a) Points R and S lie in plane V, so line RS lies in plane V.

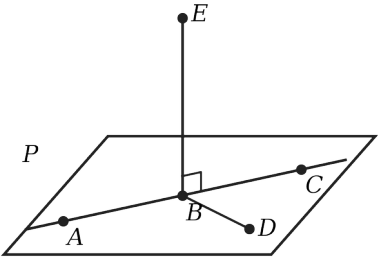
b) Lines  $m$  and  $n$  intersect at exactly one point.

c) Through points A, B, and C (noncollinear) there is exactly one plane.

d) Plane W contains at least three noncollinear points.

12. Use the diagram. Is each statement valid? Circle and explain.

- a) Points A, B, and C are collinear.  
VALID / NOT VALID
- b) Line  $EB \perp$  plane P.  
VALID / NOT VALID
- c) Line  $BD \perp$  line AC.  
VALID / NOT VALID
- d) Points B, D, and E are coplanar.  
VALID / NOT VALID



13. True or false? Explain. Two planes can intersect in exactly one point.

2.4 Algebraic Reasoning

14. Name the property that justifies each statement.

| Statement                                                                   | Property |
|-----------------------------------------------------------------------------|----------|
| $m\angle A = m\angle A$                                                     |          |
| If segment $GH \cong$ segment $JK$ , then segment $JK \cong$ segment $GH$ . |          |
| If $AB = CD$ and $CD = EF$ , then $AB = EF$ .                               |          |
| If $x = 4$ , then $3x + 1 = 3(4) + 1$ .                                     |          |
| $-7(x - 2) = -7x + 14$                                                      |          |
| If $3x = 12$ , then $x = 4$ .                                               |          |

15. Solve. Write every step and justify each one.

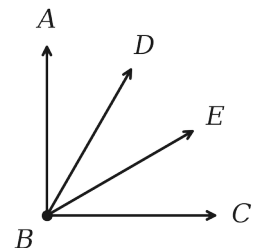
a)  $5x - 8 = 2x + 7$

b)  $-2(3y - 9) + 4y = 4$

| Statements           | Statements               |
|----------------------|--------------------------|
| 1. $5x - 8 = 2x + 7$ | 1. $-2(3y - 9) + 4y = 4$ |
| 2.                   | 2.                       |
| 3.                   | 3.                       |
| 4.                   | 4.                       |
|                      | 5.                       |

**16.** Angle Addition proof. Reasons for lines 2 and 4 are done — supply the rest.

Given:  $m\angle ABE = m\angle DBC$ .      Prove:  $m\angle ABD = m\angle EBC$ .



| Statements                     | Reasons                           |
|--------------------------------|-----------------------------------|
| 1. $m\angle ABE = m\angle DBC$ | 1. Given                          |
| 2.                             | 2. Angle Addition Postulate       |
| 3.                             | 3.                                |
| 4.                             | 4. Reflexive Property of Equality |
| 5. $m\angle ABD = m\angle EBC$ | 5.                                |

**2.5 Proving Statements about Segments and Angles**

**17.** Complete each statement.

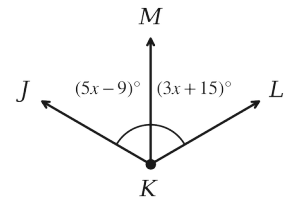
- a) Reflexive Property of Congruence: segment  $PQ \cong$  \_\_\_\_\_      b) Symmetric: If  $\angle A \cong \angle B$ , then  $\angle B \cong$  \_\_\_\_\_
- c) Transitive: If segment  $AB \cong$  segment  $CD$  and segment  $CD \cong$  segment  $EF$ , then \_\_\_\_\_

**18.** M is the midpoint of segment JK.  $JM = 9x - 4$  and  $MK = 5x + 20$ . Find x, JM, and JK.

$x =$  \_\_\_\_\_       $JM =$  \_\_\_\_\_       $JK =$  \_\_\_\_\_

**19.** Complete the proof. Statements are done — supply each reason.

Given: Ray KM bisects  $\angle JKL$ ;  $m\angle JKM = (5x - 9)^\circ$ ,  $m\angle MKL = (3x + 15)^\circ$ .  
Prove:  $x = 12$



| Statements                                                                                        | Reasons  |
|---------------------------------------------------------------------------------------------------|----------|
| 1. Ray KM bisects $\angle JKL$ ; $m\angle JKM = (5x - 9)^\circ$ , $m\angle MKL = (3x + 15)^\circ$ | 1. Given |
| 2. $\angle JKM \cong \angle MKL$                                                                  | 2.       |
| 3. $m\angle JKM = m\angle MKL$                                                                    | 3.       |
| 4. $5x - 9 = 3x + 15$                                                                             | 4.       |
| 5. $2x - 9 = 15$                                                                                  | 5.       |
| 6. $2x = 24$                                                                                      | 6.       |
| 7. $x = 12$                                                                                       | 7.       |

**20.** Overlapping segments. Statements are done — supply each reason.

Given:  $AB = CD$ .      Prove:  $AC = BD$ .



| Statements                         | Reasons  |
|------------------------------------|----------|
| 1. $AB = CD$                       | 1. Given |
| 2. $BC = BC$                       | 2.       |
| 3. $AB + BC = CD + BC$             | 3.       |
| 4. $AC = AB + BC$ ; $BD = BC + CD$ | 4.       |
| 5. $AC = BD$                       | 5.       |

**2.6 Proving Geometric Relationships**

**21.** Two lines intersect so that one of the angles measures  $48^\circ$ . Find the other three angle measures, and name the reason for each.

**22.** Find each value.

a) Vertical angles measure  $(10v - 17)^\circ$  and  $73^\circ$ . Find  $v$ .

b) A linear pair measures  $(7t + 12)^\circ$  and  $(3t - 2)^\circ$ . Find both measures.

$v =$  \_\_\_\_\_

\_\_\_\_\_  $^\circ$  and \_\_\_\_\_  $^\circ$

**23.** What can you conclude? Name the theorem.

a)  $\angle 1$  and  $\angle 2$  are both complementary to  $\angle 3$ .

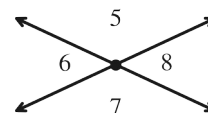
b)  $\angle 4$  and  $\angle 5$  are right angles.

c)  $\angle 6$  and  $\angle 7$  form a linear pair.

**24.** Prove the Vertical Angles Theorem. Statements are done — supply each reason.

Given:  $\angle 5$  and  $\angle 7$  are vertical angles.

Prove:  $\angle 5 \cong \angle 7$ .



| Statements                                                                                    | Reasons  |
|-----------------------------------------------------------------------------------------------|----------|
| 1. $\angle 5$ and $\angle 7$ are vertical angles                                              | 1. Given |
| 2. $\angle 5$ and $\angle 6$ form a linear pair; $\angle 6$ and $\angle 7$ form a linear pair | 2.       |
| 3. $\angle 5$ and $\angle 6$ are supplementary; $\angle 6$ and $\angle 7$ are supplementary   | 3.       |
| 4. $\angle 5 \cong \angle 7$                                                                  | 4.       |

**25.** The hard one. Congruent Complements when the shared angle is not the SAME angle — it is a congruent one. Lines 2 and 4 are done.

Given:  $\angle A$  and  $\angle B$  are complementary;  $\angle C$  and  $\angle D$  are complementary;  $\angle B \cong \angle D$ .

Prove:  $\angle A \cong \angle C$



| Statements                                                                                                             | Reasons  |
|------------------------------------------------------------------------------------------------------------------------|----------|
| 1. $\angle A$ and $\angle B$ are complementary; $\angle C$ and $\angle D$ are complementary; $\angle B \cong \angle D$ | 1. Given |
| 2. $m\angle A + m\angle B = 90^\circ$ ; $m\angle C + m\angle D = 90^\circ$                                             | 2.       |
| 3.                                                                                                                     | 3.       |
| 4. $m\angle B = m\angle D$                                                                                             | 4.       |
| 5.                                                                                                                     | 5.       |
| 6.                                                                                                                     | 6.       |
| 7.                                                                                                                     | 7.       |
| 8.                                                                                                                     | 8.       |

### REASON BANK — every reason on the test comes from this list

Given · Definition of midpoint · Definition of a segment bisector · Definition of angle bisector · Definition of congruent angles · Definition of supplementary angles · Definition of complementary angles · Definition of right angle

Segment Addition Postulate · Angle Addition Postulate · Linear Pair Postulate · Vertical Angles Theorem · Congruent Supplements Theorem · Congruent Complements Theorem · Right Angles Congruence Theorem

Distributive Property · Simplify (combine like terms) · Addition / Subtraction / Multiplication / Division Property of Equality · Substitution Property of Equality

Reflexive / Symmetric / Transitive Property of Equality · Reflexive / Symmetric / Transitive Property of Congruence